



Family Name

Given Name

Student No.

Signature

THE UNIVERSITY OF NEW SOUTH WALES

School of Electrical Engineering & Telecommunications

MID-SEMESTER EXAMINATION

Semester 2, 2018

ELEC1111

Electrical and Telecommunications Engineering

TIME ALLOWED:	75 min
TOTAL MARKS:	100
TOTAL NUMBER OF QUESTIONS:	5

THIS EXAM CONTRIBUTES 25% TO THE TOTAL COURSE ASSESSMENT

Reading Time: 5 minutes.

This paper contains 4 pages.

Candidates must **ATTEMPT ALL** questions.

Answer each question in a **separate answer booklet**.

Marks for each question are indicated beside the question.

This paper **MAY NOT** be retained by the candidate.

Print your name, student ID and question number on the front page of each answer book.

Authorised examination materials:

Candidates should use their own UNSW-approved electronic calculators.

This is a closed book examination.

Assumptions made in answering the questions should be stated explicitly.

All answers must be written in ink. Except where they are expressly required, pencils **may only be used** for drawing, sketching or graphical work.

QUESTION 1 [15 marks]

For the circuit shown in Figure 1,

- (6 marks)** Calculate the equivalent resistance R_{eq} as seen from terminals $a-b$.
- (3 marks)** Find voltage v using the result of part (a).
- (6 marks)** Use voltage division to find voltage v_1 from voltage v .

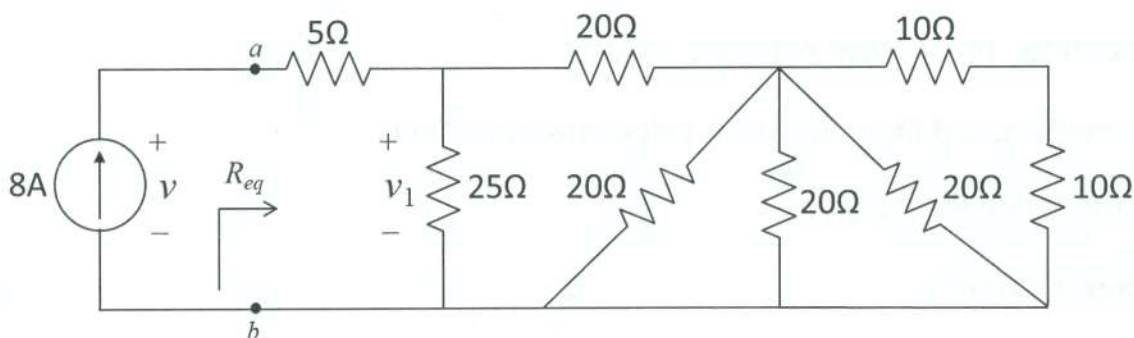
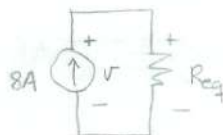


Figure 1

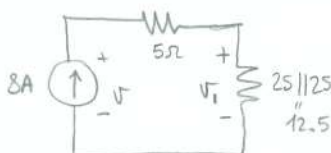
$$a) \quad R_{eq} = \left[\left[\left[(10+10) \parallel 20 \parallel 20 \parallel 20 \right] + 20 \right] \parallel 25 \right] + 5$$

$$\boxed{R_{eq} = 25 \parallel 25 + 5 = 12.5 + 5 = 17.5 \Omega}$$

$$b) \quad \boxed{V = R_{eq} \times i = 17.5 \Omega \times 8A = 140V}$$



c) We partially unsimplify the circuit and use voltage division to calculate V_1 :



$$\boxed{V_1 = V \times \frac{12.5}{17.5} = 100V}$$

QUESTION 2 [20 marks]

Use nodal analysis to calculate the power supplied/absorbed by the dependent voltage source in the circuit shown in Figure 2.

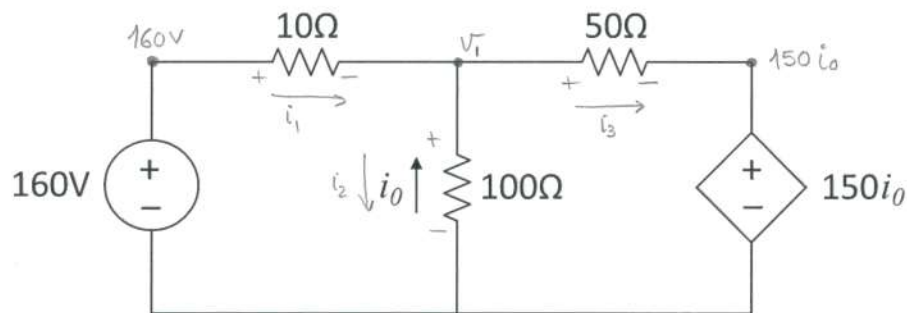


Figure 2

KCL @ node 1:

$$i_1 - i_2 - i_3 = 0$$

$$\frac{160 - V_1}{10} - \frac{V_1}{100} - \frac{V_1 - 150i_0}{50} = 0, \quad \text{where } i_0 = -i_2 = -\frac{V_1}{100}$$

Simplify equation

$$16 - \frac{V_1}{10} - \frac{V_1}{100} - \frac{V_1}{50} + \frac{150}{50} \left(\frac{V_1}{100} \right) = 0; \quad 16 - V_1 \left(\frac{1}{10} + \frac{1}{100} + \frac{1}{50} + \frac{3}{100} \right) = 0; \quad 16 - \frac{16}{100} V_1 = 0$$

We calculate the power in the dependent source:

$$\downarrow \\ \underline{\underline{V_1 = 100 \text{ V}}}$$

$$\begin{aligned} \boxed{P_{150i_0}} &= V \times i_3 = 150i_0 \times i_3 = 150 \left(\frac{-V_1}{100} \right) \times \left(\frac{V_1 - 150 \left(\frac{-V_1}{100} \right)}{50} \right) = \\ &= 150 \times (-1) \times \left(\frac{100 - 150(-1)}{50} \right) = -150 \times \frac{250}{50} = \boxed{-750 \text{ W}} \end{aligned}$$

QUESTION 3 [15 marks]

Use a series of source transformations to find i_o in the circuit shown in Figure 3.

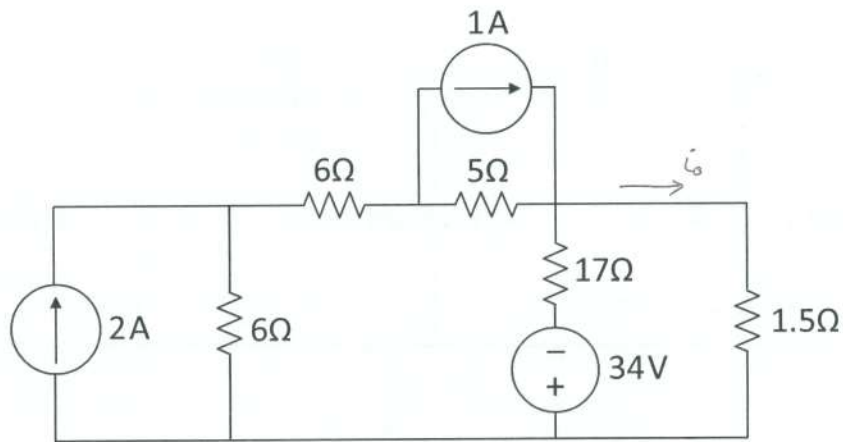
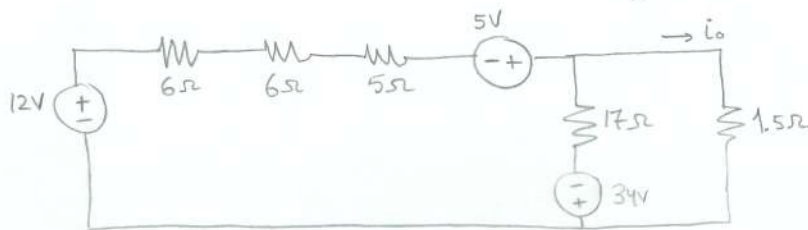


Figure 3



$$i_o = -1 \times \frac{8.5 \parallel 1.5}{1.5} = -\frac{1.275}{1.5} = -0.85 \text{ A}$$

current divider

QUESTION 4 [20 marks]

The system shown in Figure 4 is being used to power a load.

- (15 marks) Find the Thevenin equivalent of the system.
- (5 marks) Find the power in the load using your Thevenin equivalent model.

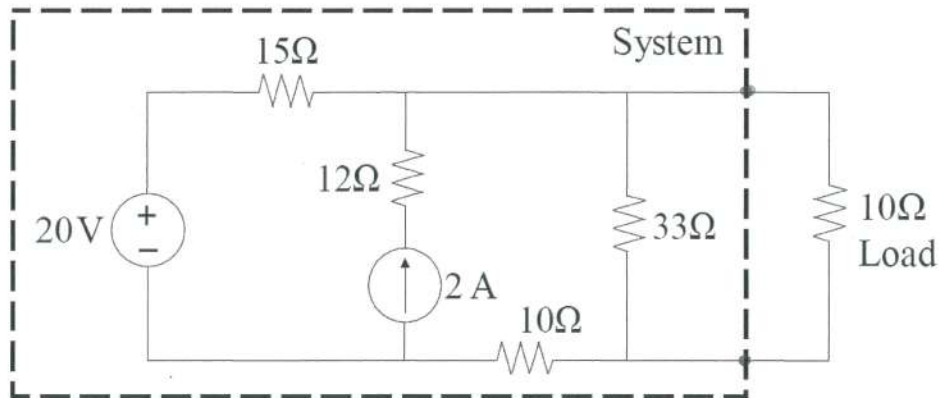
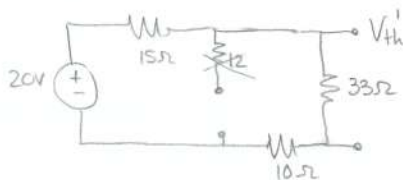


Figure 4

2) $\rightarrow V_{th}$ using superposition: (solutions for mesh and nodal analysis provided in next page).

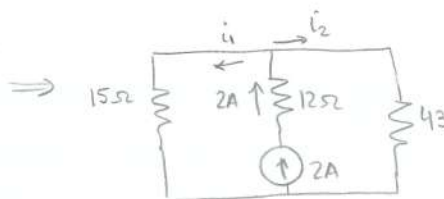
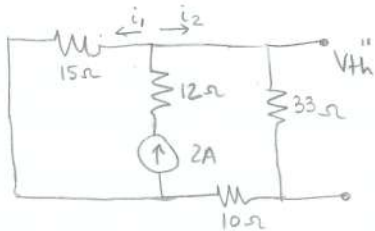
• V_{th}' due to 20V source:



$$V_{th}' = 20 \times \frac{33}{33+15+10} = 11.38 \text{ V}$$

↑
voltage divider

• V_{th}'' due to 2A source:



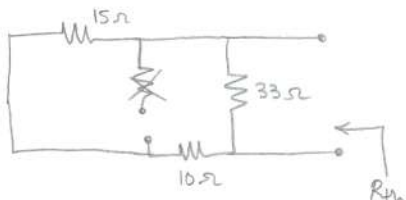
$$i_2 = 2 \times \frac{43 \parallel 15}{43} = 0.517 \text{ A}$$

↑
current divider

$$V_{th}'' = 33 i_2 = 33 \times 0.517 = 17.07 \text{ V}$$

$$\boxed{V_{th} = V_{th}' + V_{th}'' = 28.45 \text{ V}}$$

$\rightarrow R_{th}$:



$$\boxed{R_{th} = 33 \parallel 25 = 14.22 \Omega}$$

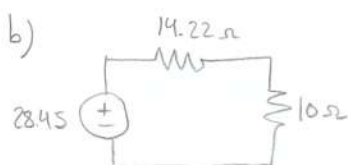
$\rightarrow$ Note. Other options to calculate R_{th} are:

* $R_{th} = \frac{V_{th}}{i_{sc}}$

* $R_{th} = \frac{V_o}{i_o}$, where $V_o = 1\text{V}$ (or $i_o = 1\text{A}$)

is a source connected to terminals a-b after turning off all the independent sources.

The value is normally 1V/1A for simplicity.



$$\boxed{P_{10\Omega} = \frac{V_{10\Omega}^2}{10} = \frac{11.74^2}{10} = 13.8 \text{ W}}$$

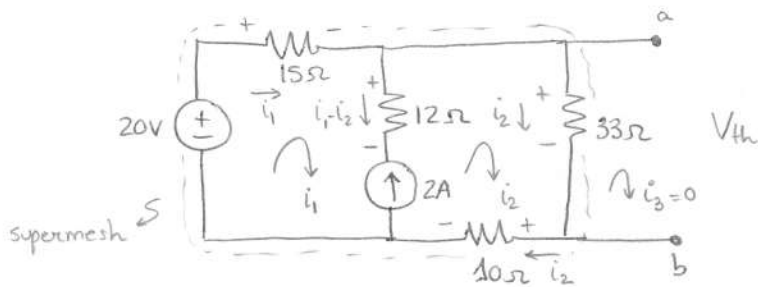
$$V_{10\Omega} = 28.45 \times \frac{10}{24.22} = 11.74 \text{ V}$$

↑
voltage divider

$$\text{OR } i = \frac{28.45}{24.22} \Rightarrow V_{10\Omega} = 10i = 11.74 \text{ V}$$

QUESTION 4

→ Solution using mesh analysis



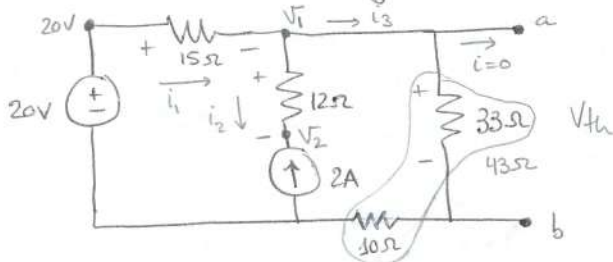
$$\left. \begin{aligned} i_1 - i_2 &= -2 \\ -20 + 15i_1 + 33i_2 + 10i_2 &= 0 \end{aligned} \right\} \begin{aligned} i_1 - i_2 &= -2 \quad (x-15) \\ 15i_1 + 43i_2 &= 20 \end{aligned} \Rightarrow \begin{aligned} -15i_1 + 15i_2 &= 30 \\ 15i_1 + 43i_2 &= 20 \end{aligned}$$

$$58i_2 = 50$$

$$\Rightarrow i_2 = \frac{50}{58} = 0.8621 \text{ A}$$

$$V_{th} = 33i_2 = 33 \times 0.8621 = 28.44 \text{ V}$$

→ Solution using nodal analysis



$$\text{KCL @ node 1: } i_1 - i_2 - i_3 = 0$$

$$\frac{20 - V_1}{15} - (-2) - \frac{V_1}{43} = 0 \Rightarrow \frac{20}{15} - \frac{V_1}{15} + 2 - \frac{V_1}{43} = 0 \Rightarrow -\left(\frac{1}{15} + \frac{1}{43}\right)V_1 = -2 - \frac{20}{15}$$

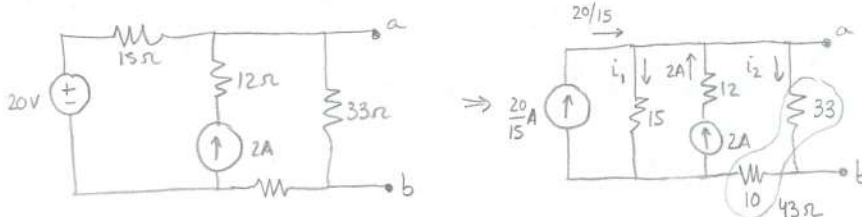
$$+0.08992V_1 = +3.333$$

$$V_1 = 37.07 \text{ V}$$

$$V_{th} = V_{33\Omega} = V_1 \frac{33}{43} = 37.07 \times \frac{33}{43} = 28.44 \text{ V}$$

voltage divider

→ Solution using source transformation



$$i_2 = \left(\frac{20}{15} + 2\right) \frac{15 \parallel 43}{43} = 3.333 \frac{11.12}{43} = 0.862 \text{ A}$$

current divider

$$V_{th} = 33i_2 = 28.44 \text{ V}$$

QUESTION 5 [30 marks]

- a. **(15 marks)** The circuit shown in Figure 5 is used to test a “touch-switch” (Sw B), which detects the touch of a finger by the capacitance of the human body. The body can be modelled as a 100 pF capacitor relative to ground, and the resistance of the arm can be modelled as a $1.5 \text{ k}\Omega$ resistor.
- (5 marks)** A person is charged to 2000 V as it walks across a carpet towards the touch-switch (Sw B). It stops in front of the touch-switch at time $t = 0 \text{ s}$ (Sw A opens). What energy is stored in the body?
 - (10 marks)** At time $t = 0 \text{ s}$, the person touches the touch-switch (Sw B closes). Calculate and plot the current through the $100 \text{ k}\Omega$ resistor for the first 50 microseconds . Ensure your plot is to scale and has at least three labelled values.

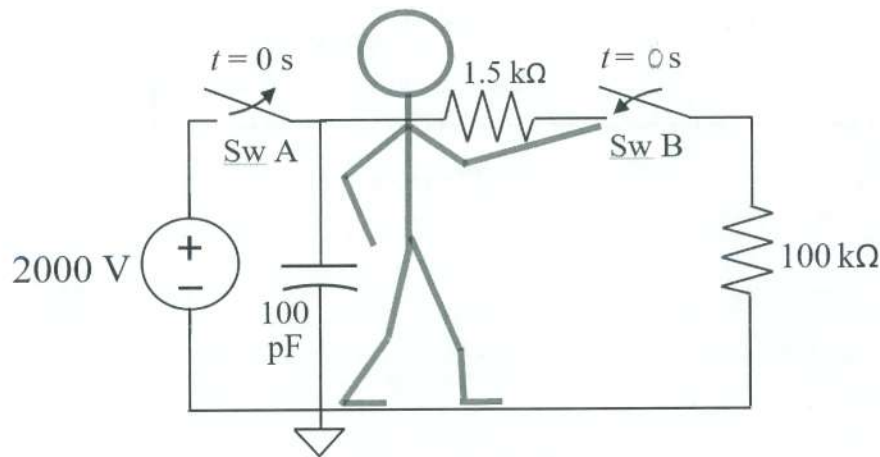


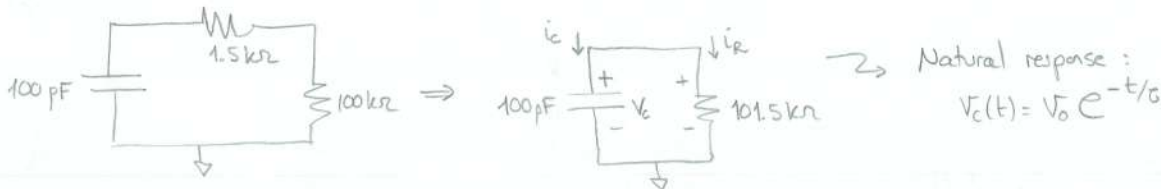
Figure 5

a)

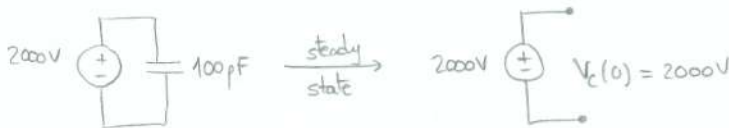
i)

$$W_c = \frac{1}{2} C V^2 = \frac{1}{2} \times 100 \times 10^{-12} \times 2000^2 = 0.2 \text{ mJ} = 200 \mu\text{J}$$

ii) $t \geq 0 \text{ s}$



- $t < 0 \Rightarrow$ we calculate initial conditions $V(1) = V_0$



- $t \geq 0 \Rightarrow$ Natural response

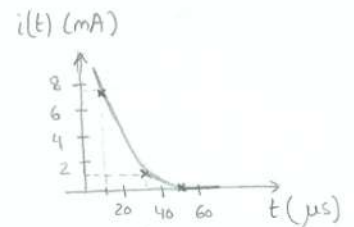
$$\tau = R \times C = 101.5 \text{ k}\Omega \times 100 \text{ pF} = 1.015 \times 10^{-5} \text{ s} = 10.15 \mu\text{s}$$

$$V_c = 2000 e^{-\frac{t}{1.015 \times 10^{-5}}} = 2000 e^{-98522t} \Rightarrow i_{100k} = \frac{V_c}{R_{eq}} = \frac{V_c}{101.5 \text{ k}\Omega} = 19.7 e^{-98522t} \text{ mA}$$

$$V(0) = V(10.15 \mu\text{s}) = 2000 e^{-1} = 735.76 \text{ V} \Rightarrow i(0) = 7.24 \text{ mA}$$

$$V(3\tau) = V(30.45 \mu\text{s}) = 2000 e^{-3} = 99.59 \text{ V} \Rightarrow i(3\tau) = 0.981 \text{ mA}$$

$$V(5\tau) = V(50.75 \mu\text{s}) = 2000 e^{-5} = 13.47 \text{ V} \Rightarrow i(5\tau) = 0.133 \text{ mA}$$



$\rightarrow$ Also: $i_{100k} = -i_c = -C \frac{dV}{dt}$

$$i_{100k} = -100 \times 10^{-12} \times 2000 \times (-98522) e^{-98522t} = 19.7 e^{-98522t} \text{ mA}$$

- b. (15 marks) In the circuit shown in Figure 6, the switch has been closed for a long time before it is opened at $t = 0$.
- (4 marks) Find the initial voltage $v_c(0^-)$ across the capacitor.
 - (4 marks) Find the final voltage $v_c(\infty)$ across the capacitor.
 - (4 marks) Derive an expression for the voltage of the capacitor $v_c(t)$ for all time (i.e., for both $t < 0$ and $t \geq 0$).
 - (3 marks) Sketch the voltage $v_c(t)$ obtained in part (iii) as a function of time.

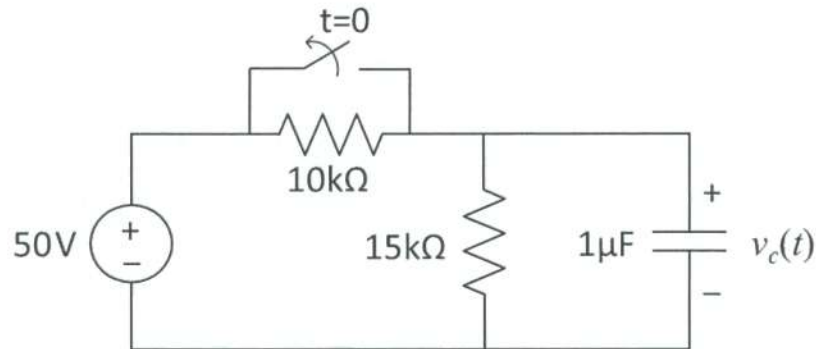


Figure 6

END OF PAPER

i) $t < 0$



$$v_c(0^-) = 50 \text{ V}$$

Capacitor behaves as open circuit in DC steady state

ii) $t \geq 0$



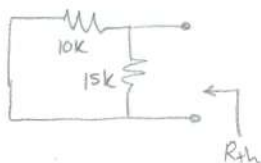
$$v_c(\infty) = 50 \frac{15 \text{ k}\Omega}{10 \text{ k}\Omega + 15 \text{ k}\Omega} = \frac{50 \times 15}{25} = 30 \text{ V}$$

↑
voltage divider

iii)

$$v_c(t) = v(\infty) + (v(0^-) - v(\infty)) e^{-t/\tau}$$

where $\tau = R_{th} C$ in the circuit in part ii)



$$R_{th} = 10 \text{ k}\Omega \parallel 15 \text{ k}\Omega = 6 \text{ k}\Omega$$

$$\tau = 6 \text{ k}\Omega \times 1 \mu\text{F} = 6 \text{ ms}$$

$$v_c(t) = 30 + (50 - 30) e^{-t/6 \text{ ms}} = 30 + 20 e^{-t/6 \text{ ms}}$$

$$v_c(t) = \begin{cases} 50 \text{ V}, & t < 0 \\ 30 + 20 e^{-t/6 \text{ ms}}, & t \geq 0 \end{cases}$$

iv) $v_c(t) \text{ (V)}$

